

FOZ DO IGUACU, Brazil

WMO: 838270

Lat: 25.596S

Lon: 54.487W

Elev: 240

StdP: 98.48

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 3.8	(c) 5.9	(d) 0.8	(e) 4.1	(f) 8.1	(g) 2.8	(h) 4.8	(i) 9.0	(j) 9.1	(k) 24.1	(l) 8.2	(m) 23.0	(n) 1.5	(o) 190	(p) 0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	35.2	22.9	34.2	23.0	33.2	23.1	26.1	31.6	25.6	31.0	25.0	30.4	4.3	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	19.9	28.7	24.0	19.5	28.3	23.3	18.7	27.6	82.1	31.7	79.8	31.2	77.6	30.4	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.2	7.1	6.1		-0.3	37.6	1.9	1.2	-1.6	38.5	-2.7	39.2	-3.8	39.8	-5.2	40.7	
				-0.5	27.3	1.9	0.8	-1.9	27.8	-3.0	28.3	-4.1	28.7	-5.4	29.2	
		DB														
		WB														

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.1	26.4	26.1	25.0	22.4	18.6	17.5	17.1	19.2	20.6	23.0	24.0	25.6
	DBStd	4.80	1.84	2.05	2.38	3.58	3.78	4.39	4.70	4.83	4.50	3.23	2.58	2.18
	HDD10.0	12	0	0	0	0	1	3	6	2	1	0	0	0
	HDD18.3	286	0	0	1	9	44	68	79	52	28	3	1	0
	CDD10.0	4427	509	450	464	373	267	227	226	287	318	402	421	483
	CDD18.3	1658	251	217	207	133	52	42	42	78	96	146	171	224
	CDH23.3	14197	2338	1889	1677	972	291	231	286	691	896	1273	1573	2079
(14) Wind	WSAvg	2.7	2.5	2.5	2.4	2.4	2.4	2.6	2.7	3.0	3.1	3.1	2.8	2.5
	PrecAvg	1767	164	143	125	136	163	131	103	89	152	215	160	185
	PrecMax	2553	369	299	327	328	404	307	240	341	325	605	341	457
	PrecMin	1247	17	11	42	24	0	11	27	16	41	43	31	22
	PrecStd	326	73	66	75	76	112	72	54	82	72	110	83	108
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	37.0	35.8	33.8	31.1	29.2	30.0	33.1	36.0	36.1	35.8	36.0
		MCWB	24.0	22.3	22.8	23.5	22.5	21.2	19.6	19.6	20.8	23.1	23.0	22.9
	2%	DB	34.2	34.5	33.9	32.1	28.2	27.9	28.2	31.6	33.8	34.0	33.9	34.1
		MCWB	24.0	23.7	23.0	23.4	21.2	20.2	19.2	19.3	20.2	22.5	22.4	23.5
	5%	DB	33.1	33.0	32.1	30.5	26.6	26.1	26.8	29.9	31.2	32.0	32.1	33.0
		MCWB	24.1	23.9	23.2	22.7	20.7	19.8	19.0	19.2	20.2	21.9	22.0	23.4
	10%	DB	32.0	31.4	30.8	28.9	24.7	24.1	24.8	27.7	28.8	29.9	30.8	31.6
		MCWB	23.9	23.6	22.9	22.3	20.0	19.5	18.2	18.7	19.6	21.3	21.8	23.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	26.5	26.0	25.5	24.2	22.8	21.7	22.4	23.8	25.2	25.7	26.4
		MCDB	32.3	31.6	31.3	31.0	28.6	26.6	26.7	29.3	31.4	32.8	31.8	32.0
	2%	WB	26.0	25.7	25.1	24.5	22.6	21.7	20.7	21.0	22.3	24.1	24.7	25.7
		MCDB	31.6	30.8	30.7	29.7	26.4	25.8	25.7	27.9	28.7	31.1	30.5	31.2
	5%	WB	25.5	25.1	24.5	23.8	21.6	20.7	19.7	20.1	21.3	23.1	23.7	25.1
		MCDB	30.9	30.0	29.6	28.6	24.9	24.6	24.5	26.5	27.3	29.3	29.4	30.3
	10%	WB	24.8	24.5	23.8	22.9	20.6	19.7	18.8	19.3	20.4	22.3	22.9	24.4
		MCDB	30.0	29.1	28.4	27.1	23.7	23.1	23.4	25.5	26.1	27.5	28.1	29.1
(36) Mean Daily Temperature Range	5% DB	MDBR	10.9	10.7	10.9	10.6	9.8	9.8	11.1	12.5	12.2	11.3	11.8	11.2
		MCDBR	12.6	12.8	12.8	12.1	11.5	11.5	12.8	14.7	15.5	14.5	14.3	13.5
		MCWBR	4.8	4.6	4.7	4.9	5.5	5.5	5.7	5.6	5.7	5.6	5.6	5.2
	5% WB	MCDBR	10.6	10.5	10.6	10.5	9.6	9.8	11.1	12.9	12.3	12.1	11.2	10.8
		MCWBR	4.9	4.8	5.0	5.1	5.2	5.5	5.8	6.2	5.8	5.6	5.4	5.3
(40) Clear-Sky Solar Irradiance	taub		0.404	0.398	0.388	0.383	0.356	0.389	0.385	0.468	0.565	0.471	0.419	0.415
	taud		2.430	2.445	2.464	2.448	2.493	2.398	2.386	2.108	1.859	2.190	2.341	2.378
	Ebn at Noon		940	930	910	868	850	790	809	769	738	853	920	932
	Edh at Noon		124	119	111	104	91	95	100	144	201	152	135	131
(44) All-Sky Solar Radiation		RadAvg	6.44	5.89	5.38	4.44	3.29	2.85	3.22	3.97	4.53	5.18	6.31	6.46
		RadStd	0.51	0.48	0.36	0.41	0.46	0.30	0.39	0.29	0.48	0.59	0.63	0.63

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page